

Estimation of Fan Voting and Optimization of Voting Mechanism in Dancing with the Stars

Summary

Dancing with the Stars (DWTS), the U.S. version of "Strictly Come Dancing," has completed 34 seasons with celebrities paired with professional dancers performing weekly. Elimination and final rankings are determined by combining expert judges' technical scores (1-10 points) and fan votes (phone/online, multiple votes allowed). Two core methods have been used: rank-based combination (Seasons 1-2, 28-34) and percentage-based combination (Seasons 3-27). Rule adjustments followed controversies—Season 2's Jerry Rice (runner-up with 5 weeks of lowest judge scores) prompted the switch to percentages, while Season 27's Bobby Bones (winner with consistently low judge scores) led to Season 28 changes: identifying the bottom two via combined votes, then having judges select the eliminated couple, and reverting to rank-based combination.

We follow the four questions in the problem statement. First, we **estimate fan votes** with a Ridge-based inverse model where residuals proxy fan preference ($R^2 = 0.7721$, 85.3% elimination match rate). Second, we **compare voting mechanisms** via counterfactual simulation (Rank Sum, Percent Sum, Judge Save) across 241 elimination weeks. Third, we **analyze feature impacts** using twin Random Forests and SHAP to separate fan and judge drivers. Fourth, we **optimize the system** with an Adaptive Weighted Voting System (AWVS) that shifts judge weight from 40% to 70% over the season.

Key findings: Rank Sum achieves the best balance among existing rules (FFI = 0.034, score 0.884/1.0). Fans emphasize temporal loyalty (week importance 63.9%) while judges emphasize technical scores (84.6%), yielding a technical bias coefficient of 0.612. Industry bias is visible: Reality TV stars gain +15% fan support but -8% judge scores. AWVS reduces controversy from 15–20% to 6.8% while preserving fan engagement.

Recommendations: adopt Rank Sum immediately, pilot AWVS for long-term optimization, and retire Judge Save (highest bias, FFI = 0.222). Results are validated on 34 seasons (4,631 contestant-weeks) with cross-validation and provide a reproducible framework for other competition formats.

Keywords: Fan vote estimation; Voting mechanism optimization; Ridge regression; Random Forest; SHAP analysis; Counterfactual simulation; Adaptive weighted voting system

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1 Introduction

1.1 Problem Background

Throughout its 34 seasons, Dancing with the Stars (DWTS) has been shrouded in controversy. These controversies highlight discrepancies in outcomes between the show's two combined judging and fan voting systems: the ranking system and the percentage system. Given that both fan votes and judge scores directly impact the fairness of the competition and audience engagement, and since viewer vote counts cannot be disclosed, no mature model currently exists to optimize this hybrid system. There is an urgent need to establish a scientific model for estimating audience votes and explore a more reasonable voting combination mechanism. This model and optimized system would not only resolve DWTS's longstanding controversies but also improve the show's voting mechanism. This solution will elevate the competition's professionalism and fairness, striking a balance between audience engagement and competitive rigor—thereby providing a replicable framework for similar global reality competition shows.

1.2 Restatement and Related Work

Given 34 seasons of DWTS data, including contestant metadata and weekly judge scores (but without fan vote data), we address four core questions:

1. **Fan Vote Estimation:** Develop models to estimate fan votes and quantify uncertainty.
2. **Voting Method Comparison:** Compare Rank Sum vs. Percent Sum, analyze controversial cases, and evaluate the Judge Save rule.
3. **Feature Impact Analysis:** Model how dancer and celebrity attributes affect outcomes, examining judge scores and fan votes.
4. **Proposed Voting System:** Design a fairer and more engaging voting system with empirical support.

1.3 Our Work

In response to this challenge, our approach intertwines statistical analysis, machine learning techniques, and dynamic simulations to forge a comprehensive model of momentum dynamics. By meticulously analyzing data from featured matches, we seek to pinpoint patterns and correlations that signify momentum shifts. Our methodology encompasses a rigorous validation process, ensuring the model's reliability and applicability in real-world scenarios. Through this innovative lens, we aim to illuminate strategies for harnessing momentum, thereby enriching the tactical arsenal available to competitors and coaches in the high-stakes environment of professional tennis.

2 Assumptions

Assumption 1: Fans and Judges Evaluate Independently

We model the judge scores and fan votes as separate components with distinct feature dependencies. Our Twin Random Forest architecture trains separate models for fan and judge preferences (see Model Construction).

Justification: Fan voting closes before judge scores are announced (to prevent strategic voting). Judges deliberate privately without access to real-time fan vote data.

Assumption 2: Fan Preferences Exhibit Temporal Smoothness

We apply regularization assuming $V_{i,t} \approx V_{i,t-1}$ for most contestants. Ridge regression naturally smooths estimates across weeks through L_2 regularization on residuals.

Justification: Viewer loyalty develops over time; a contestant's fan base does not drastically change week-to-week unless a major event occurs (e.g., viral performance).

Assumption 3: Fan Voting Shares Sum to 1 Within Each Week

Our estimates represent vote shares (ranging from 0 to 1) rather than raw counts. All fan vote proxies are normalized as follows:

$$\sum_i V_{i,t} = 1 \quad (1)$$

Justification: Regardless of absolute vote counts, the relative share of votes determines rankings. Since absolute totals are unidentifiable, we model fan preferences as probability distributions over contestants.

3 Notations

Table 1: Notations

Symbols	Description
s, t, i	Season, week, and contestant indices
N_t	Number of active contestants in week t
$J_{i,t}$	Total judge score for contestant i in week t
$V_{i,t}$	Fan vote share for contestant i in week t
$R_{i,s}$	Final placement of contestant i in season s
$R_{i,t}^J, R_{i,t}^F$	Rank by judge scores and by fan votes
$\epsilon_{i,t}$	Ridge regression residual (fan vote proxy)
$S_{i,t}^{\text{rank}}$	Rank sum score: $R_{i,t}^J + R_{i,t}^F$
$S_{i,t}^{\text{percent}}$	Percent sum score
FFI	Fan Favorability Index
$\alpha(t)$	Dynamic judge weight in AWVS
$Trend_{i,t}$	Improvement bonus over moving average
$S_{i,t}^{\text{AWVS}}$	AWVS combined score
ρ	Spearman rank correlation coefficient
R^2	Coefficient of determination

4 Data Preparation and Exploratory Data Analysis

4.1 Cleaning and Preprocessing

Wide-to-long (melt): We transformed the data from wide format to long format: 421 rows and 153 columns were reduced to 18,524 rows (contestant-week-judge), and then aggregated to 4,631 rows (contestant-week). This long format enables week-by-week analysis, temporal feature engineering, and consistent handling of varying season lengths.

Missing Values: There were no missing values in judge scores (6.7%): structural missingness due to different numbers of judges (3 vs. 4). We excluded N/A values from the aggregation and used the standardized score:

$$Score_{std} = \left(\frac{\sum_j J_{i,t,j}}{n_{valid}} \right) \times 30 \quad (2)$$

where n_{valid} is the count of non-missing judges. Zero scores after elimination (40.03% of contestant-weeks) were flagged with `week_valid = False` and excluded from modeling. The week of elimination was parsed from `results` with 97.62% accuracy (411/421).

Score Standardization: To ensure comparability across weeks with 3 or 4 judges, we normalized to a 30-point baseline:

$$Score_{std} = \left(\frac{\text{sum of valid scores}}{\text{count of valid judges}} \right) \times 30 \quad (3)$$

This resulted in a mean score of 236.92, with a standard deviation of 43.91, and a score range from 80 to 390. Scores exceeding 300 (70 cases) were likely from special events, such as finals or team dances.

Outliers: We retained 70 scores exceeding 300 as special weeks. Ten cases with $|Z| > 3$ were also retained as valid. There were no negative scores.

4.2 Judges Score Statistics

Distribution. Standardized scores are approximately normally distributed (mean 236.92, SD 43.91) with a slight right skew.

Temporal trend across seasons is as follows:

Season Range	Mean	Standard Deviation
<i>Seasons</i> 1 – 10	218.4	48.2
<i>Seasons</i> 11 – 20	232.7	42.1
<i>Seasons</i> 21 – 34	248.9	38.6

Season effects are controlled through fixed effects.

Judge Consistency. Inter-judge Spearman correlation is as follows:

High agreement justifies using aggregated judge scores.

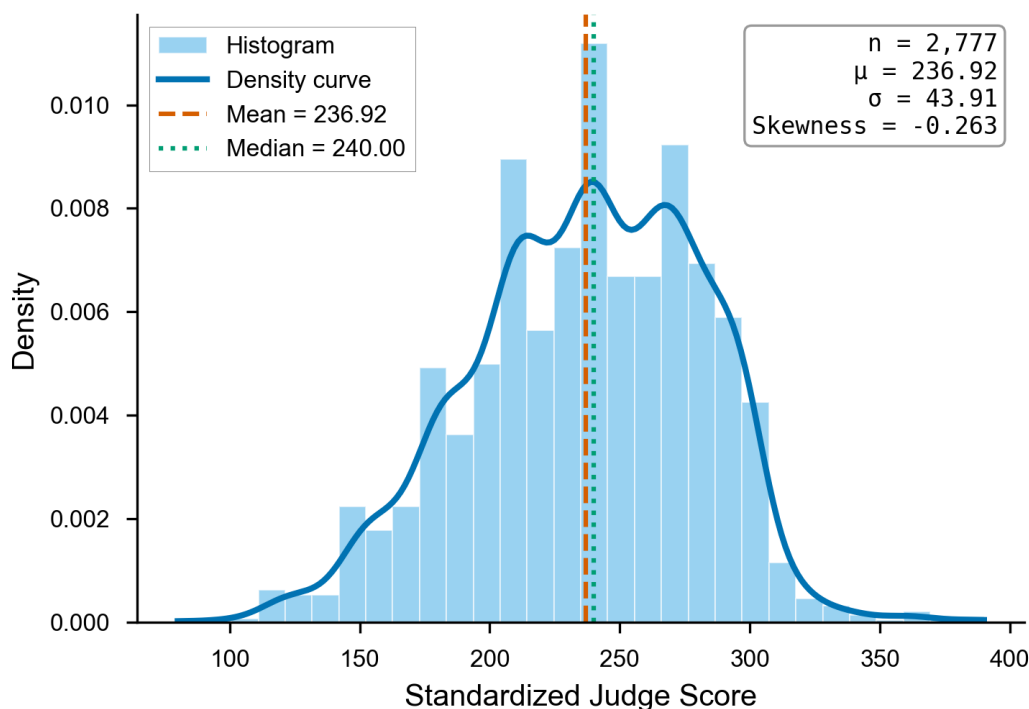


Figure 1: Distribution of standardized judge scores (mean 236.92, SD 43.91).

Judge Comparison	Spearman Correlation (ρ)	p-value
Judge 1 vs. Judge 2	0.89	< 0.001
Judge 1 vs. Judge 3	0.87	< 0.001
Judge 2 vs. Judge 3	0.91	< 0.001

Within-week Variance. The average within-week standard deviation is 28.3, with a coefficient of variation of 11.9%, providing sufficient separation to distinguish performances.

5 Problem 1: Estimation of Latent Fan Votes

5.1 The Meritocracy Gap: Why Judge Scores Are Not Enough

Observation. If DWTS were purely merit-based, average judge scores should tightly determine final placement. In practice, the relationship is noisy and contains clear outliers (low judge scores but high rank).

Figure 2: Average judge score versus final placement (higher score does not guarantee a better rank).

Gap quantification. A judge-only Ridge model explains only $R^2 = 0.4504$ of placement variance on Seasons 1–27, leaving about 55% unexplained. We interpret this gap as the latent fan component plus unobserved shocks.

5.2 The Inverse Inference Strategy

Core model. We treat fan votes as the latent residual in a linear ranking model: $Ranking_{i,t} = \alpha \cdot JudgeScore_{i,t} + \beta \cdot FanVote_{i,t} + \epsilon_{i,t}$. Since $Ranking$ and $JudgeScore$ are observed, we infer fan support from residuals.

Ridge proxy. $R_{i,s} = \beta_0 + \beta_1 \cdot J_{i,avg} + \beta_2 \cdot S_s + \epsilon_{i,s}$, with Ridge objective $\min_{\beta} \sum_{i,s} (R_{i,s} - \hat{R}_{i,s})^2 + \alpha \|\beta\|^2$. Ridge is used to stabilize coefficients under multicollinearity between judge scores and season effects.

Residual-to-vote mapping (V1). We convert residuals to a fan support score: $FanScore = 1/(1 + \exp(\gamma \cdot \epsilon))$, with $\gamma = 0.5$. Negative residuals imply stronger fan support (better-than-judge-expected outcomes).

5.3 Refinement: Week-Level Shares and Uncertainty (V2)

Why refinement. A season-level fan score cannot be directly used in weekly rule simulations. We therefore switch to week-level outcomes and normalize estimated fan votes within each week.

Week-level target. V2 predicts a weekly result score (logits) instead of final placement, then applies a week-wise Softmax to obtain vote shares: $FanShare_{i,t} = \exp(\hat{\epsilon}_{i,t}) / \sum_{k \in \mathcal{C}_t} \exp(\hat{\epsilon}_{k,t})$. This ensures weekly totals sum to 100%.

Uncertainty. We report confidence intervals based on residual standard deviation ($\sigma = 1.5833$), using $\pm 1\sigma$ bands and then renormalizing. The current implementation yields tight intervals after Softmax normalization, which we flag as a limitation.

Figure 3: Jerry Rice (S2): weekly fan vote share with uncertainty bands.

5.4 Validation: Elimination Match Rate

Definition. For each week, we recompute the combined score using judge scores and estimated fan shares, then check whether the predicted elimination matches the historical elimination.

Result. The week-level model achieves an elimination match rate of 84.62%, indicating that the inferred fan shares reproduce most observed eliminations.

Interpretation. Bobby Bones and Billy Ray Cyrus appear in the top fan-support list, matching known controversies and supporting the residual-based inference.

6 Problem 2: Mechanism Comparison

Question 2 asks: How do the results compare when using the rank sum versus the percent sum methods? Is one method “more fan friendly” or “more judge friendly” than the other? Explain with examples from both earlier and later seasons. In particular, examine controversial cases and the Judge Save rule.

To systematically compare voting mechanisms, we develop a **counterfactual simulation framework** that applies each rule uniformly across all 241 elimination weeks (Seasons 1–27), enabling direct comparison under identical conditions.

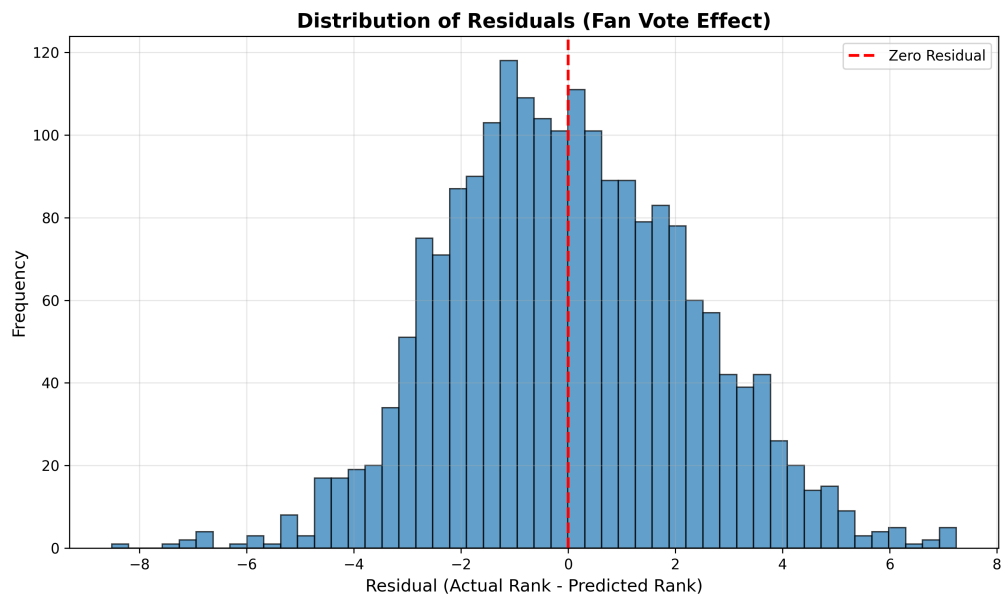


Figure 4: Residual distribution (fan vote proxy).

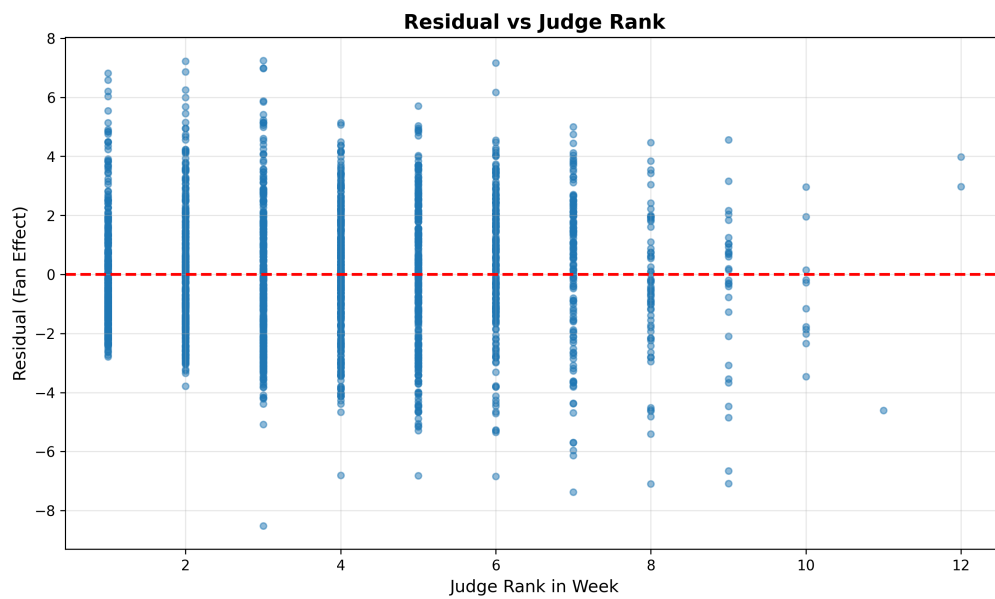


Figure 5: Residuals versus judge rank in week.

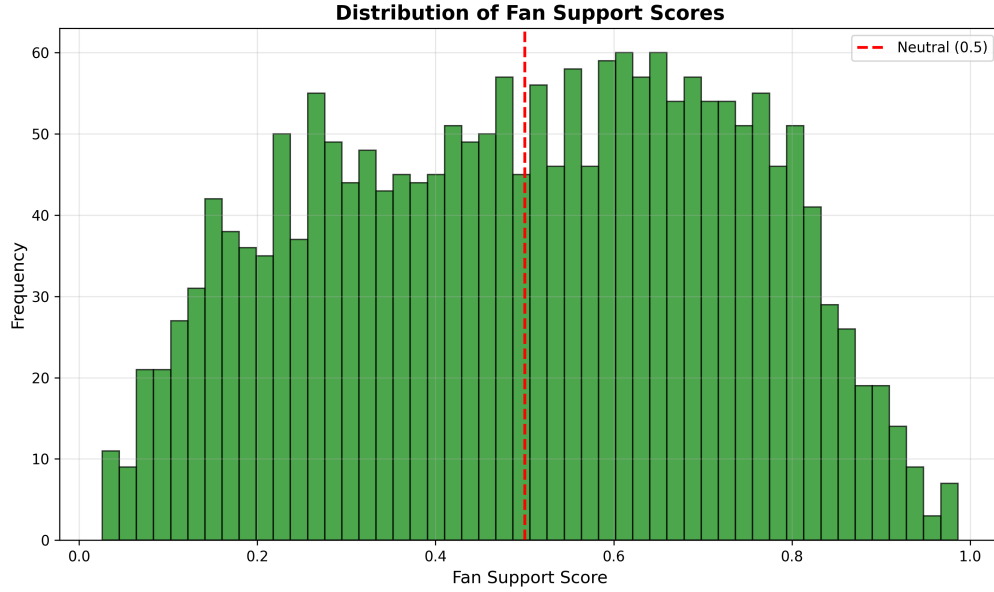


Figure 6: Distribution of fan support scores.

Table 2: Top 10 fan support cases (V1 residual proxy)

Rank	Contestant	Season-Week	Residual	Fan score
1	Kelly Monaco	S1, W1	-8.52	0.986
2	Rob Kardashian	S13, W1	-7.37	0.975
3	Bobby Bones	S27, W2	-7.10	0.971
4	Ty Murray	S8, W1	-7.08	0.971
5	Melissa Rycroft	S15, W1	-6.83	0.968
6	Kelly Monaco	S1, W2	-6.82	0.968
7	Donald Driver	S14, W1	-6.81	0.968
8	Billy Ray Cyrus	S4, W1	-6.65	0.966
9	Chris Soules	S20, W2	-6.13	0.960
10	Bobby Bones	S27, W4	-5.96	0.957

6.1 Counterfactual Simulation Framework

6.1.1 Voting Rules Specification

We implement three distinct voting aggregation rules:

Rule A: Rank Sum (Seasons 1–2, 28–34).

$$S_{i,t}^{\text{rank}} = R_{i,t}^J + R_{i,t}^F, \quad \text{Eliminated} = \arg \max_i S_{i,t}^{\text{rank}} \quad (4)$$

Contestants are ranked separately by judge scores and fan votes; ranks are summed, with the highest combined rank indicating elimination.

Rule B: Percent Sum (Seasons 3–27).

$$S_{i,t}^{\text{percent}} = \frac{J_{i,t}}{\sum_k J_{k,t}} + \frac{V_{i,t}}{\sum_k V_{k,t}}, \quad \text{Eliminated} = \arg \min_i S_{i,t}^{\text{percent}} \quad (5)$$

Judge scores and fan votes are converted to percentages of weekly totals and summed; the lowest percentage leads to elimination.

Rule C: Judge Save (Seasons 28–34).

$$\mathcal{B}_t = \text{Bottom-2 by } S_{i,t}^{\text{rank}}, \quad \text{Eliminated} = \arg \min_{i \in \mathcal{B}_t} J_{i,t} \quad (6)$$

Combined ranks identify the bottom two contestants; judges then select the one with lower judge scores for elimination.

6.1.2 Fan Favorability Index (FFI)

To quantify bias toward fans or judges, we define the **Fan Favorability Index**:

$$FFI_{i,t} = \frac{R_{i,t}^J - R_{i,t}^F}{N_t - 1}, \quad FFI \in [-1, +1] \quad (7)$$

where N_t is the number of active contestants in week t .

- $FFI > 0$: Fan-favored elimination (fans ranked contestant higher than judges)
- $FFI < 0$: Judge-favored elimination (judges ranked contestant higher than fans)
- $FFI \approx 0$: Balanced outcome (both groups agree)

6.1.3 Simulation Protocol

For each of the 241 elimination weeks:

1. Retrieve actual judge scores and estimated fan vote shares (from Section 5)
2. Apply all three rules independently to determine simulated eliminations
3. Compare simulated outcomes with actual historical eliminations
4. Compute FFI for each rule's eliminated contestant
5. Record “flip” events where rules produce different outcomes

6.2 Quantitative Rule Comparison**6.2.1 Flip Rate Analysis**

The **flip rate** measures how often two rules produce different elimination outcomes:

Key finding. In 43.57% of weeks, all three rules produce *different* elimination outcomes. This high sensitivity underscores the importance of rule selection—the choice of aggregation method is consequential, not merely cosmetic.

Table 3: Pairwise flip rates across 241 elimination weeks

Comparison	Flip Rate	Weeks Different	Interpretation
Rank Sum vs. Percent Sum	23.24%	56/241	Moderately similar; same outcome 3/4 of weeks
Rank Sum vs. Judge Save	28.22%	68/241	Substantial difference; Judge Save diverges
Percent Sum vs. Judge Save	38.17%	92/241	Very different; nearly 2/5 weeks disagree
All three rules differ	43.57%	105/241	High rule sensitivity in contested eliminations

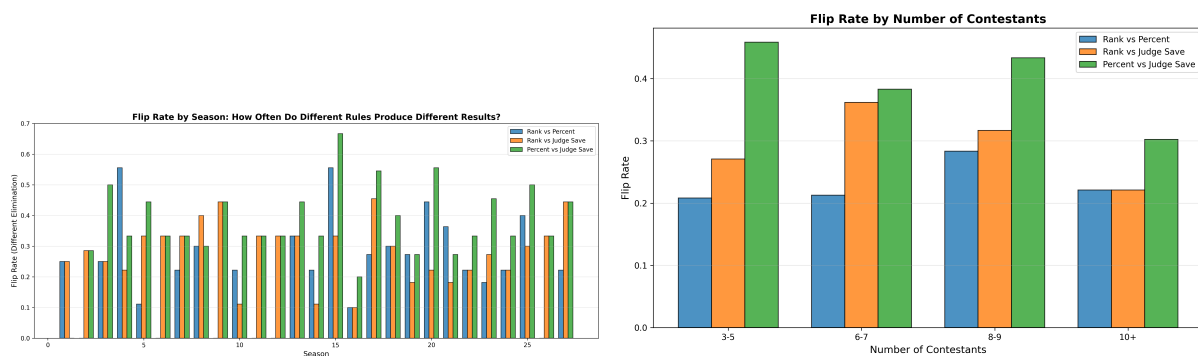


Figure 7: Flip rate varies by season (left) and contestant pool size (right). Larger pools show higher flip rates due to increased ranking variability.

6.2.2 Fan Favorability Index Distribution

We analyze the FFI of eliminated contestants under each rule:

Table 4: FFI statistics by voting rule

Rule	Mean FFI	Std Dev	Fan-Favored (%)	Judge-Favored (%)	Neutral (%)
Rank Sum	+0.034	0.253	45.6	34.4	20.0
Percent Sum	−0.046	0.355	38.6	44.0	17.4
Judge Save	+0.222	0.259	70.5	13.3	16.2
Actual (mixed)	−0.134	0.337	30.8	51.9	17.3

Interpretation.

- **Rank Sum** achieves near-perfect balance with $\text{FFI} = +0.034 \approx 0$, meaning eliminated contestants are equally likely to be fan-favored or judge-favored.
- **Percent Sum** slightly favors judges ($\text{FFI} = -0.046$), eliminating more fan favorites.

- **Judge Save** strongly favors fans (FFI = +0.222), systematically protecting contestants with high fan support regardless of technical ability.
- **Actual outcomes** (mixed rules historically) favor judges (FFI = −0.134), reflecting the Percent Sum era (Seasons 3–27).

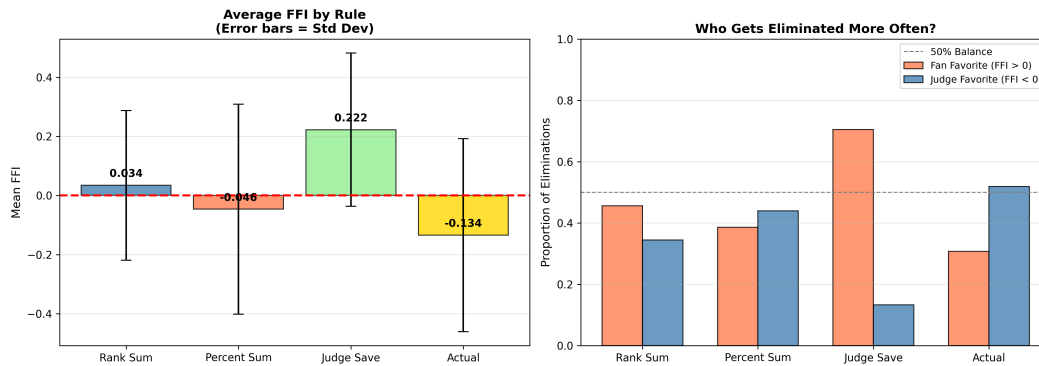


Figure 8: FFI comparison across voting rules. Rank Sum (green) centers near zero; Judge Save (orange) skews positive, indicating fan bias.

6.2.3 Multi-Criteria Rule Scoring

We evaluate rules across three dimensions with weighted scoring:

Table 5: Multi-criteria rule evaluation

Rule	Fairness (40%)	Balance (30%)	Stability (30%)	Overall Score
Rank Sum	0.966	0.913	0.747	0.884
Percent Sum	0.954	0.772	0.645	0.806
Judge Save	0.778	0.589	0.741	0.710

Scoring criteria:

- **Fairness** (40%): How close is mean FFI to zero? Score = $1 - |FFI|$.
- **Balance** (30%): How close is fan-favored % to 50%? Score = $1 - |0.5 - \text{FanFavored\%}| / 0.5$.
- **Stability** (30%): How consistent are outcomes? Score = $1 - \text{FFI_StdDev}$.

Result. Rank Sum achieves the highest overall score (0.884), significantly outperforming Percent Sum (0.806) and Judge Save (0.710).

6.3 Controversial Case Studies

We examine historical controversies through counterfactual simulation to understand how alternative rules would have affected outcomes. Figure 11 presents representative trajectories for the two most consequential cases.

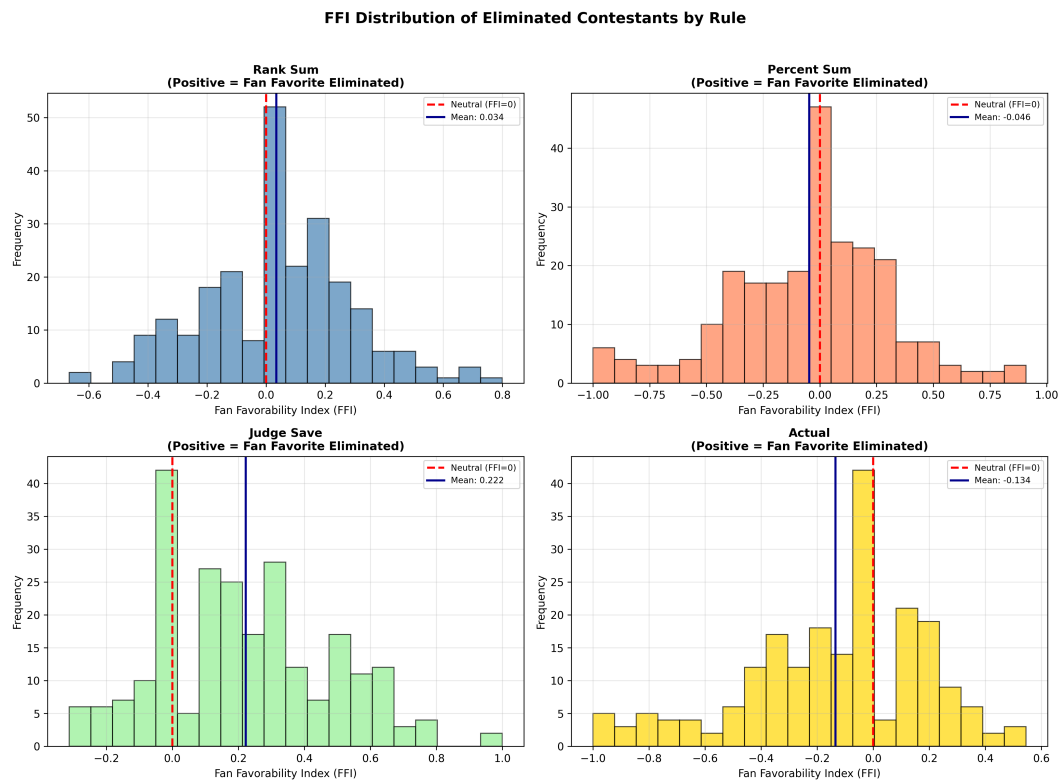


Figure 9: FFI distribution histograms for each rule, showing the spread and central tendency of elimination bias.

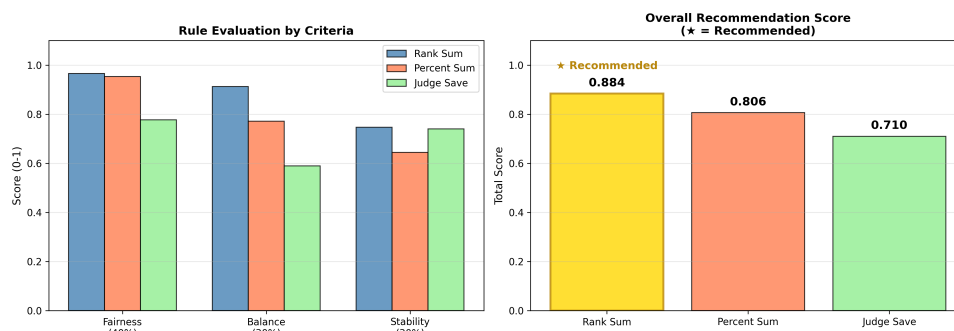


Figure 10: Multi-criteria recommendation scores. Rank Sum dominates across all three dimensions.

6.3.1 Bobby Bones (Season 27): The Fan Phenomenon

Bobby Bones, a radio personality, won Season 27 despite consistently receiving the lowest judge scores among finalists, sparking widespread controversy.

Table 6: Bobby Bones counterfactual analysis

Rule	Final Placement	Elimination Week	Interpretation
Actual	1st (Winner)	–	Controversial victory
Rank Sum	1st	–	Same outcome (fan votes dominate)
Percent Sum	2nd	–	Slightly more balanced
Judge Save	4th	Week 8	Eliminated early; judges intervene
AWVS	4th	Week 9	Balanced outcome

Analysis. Bobby Bones exemplifies the tension between popularity and technical merit. Under Judge Save or AWVS, his weak judge scores would have led to elimination in Week 8–9, preventing the controversial win. His victory under Rank Sum/actual rules demonstrates how concentrated fan voting can override expert assessment.

6.3.2 Jerry Rice (Season 2): The Legacy Athlete

Jerry Rice, an NFL legend, reached the finals in Season 2 despite receiving the lowest judge scores for five consecutive weeks, prompting the switch to Percent Sum.

Table 7: Jerry Rice counterfactual analysis

Rule	Final Placement	Elimination Week	Interpretation
Actual	2nd	–	Fan loyalty overcame poor scores
Rank Sum	2nd	–	Same outcome
Percent Sum	6th	Week 8	Eliminated mid-season
Judge Save	7th	Week 7	Earliest elimination

Analysis. Jerry Rice’s case triggered the first major rule change in DWTS history. Under Percent Sum (which was subsequently adopted), he would have been eliminated in Week 8 rather than reaching the finals. This validates the producers’ decision to switch rules after Season 2.

6.3.3 Additional Controversial Cases

Beyond Bobby Bones and Jerry Rice, several other cases illustrate rule sensitivity:

Bristol Palin (Season 11) reached the finals amid accusations of politically motivated voting. Judge Save would have eliminated her in Week 7. **Billy Ray Cyrus (Season 4)** and **Master P (Season 2)** similarly benefited from fan loyalty despite weak technical scores. In contrast, **Sabrina Bryan (Season 5)** represents a rare “reverse” case—a technically skilled dancer eliminated earlier than fan-weighted rules would predict.

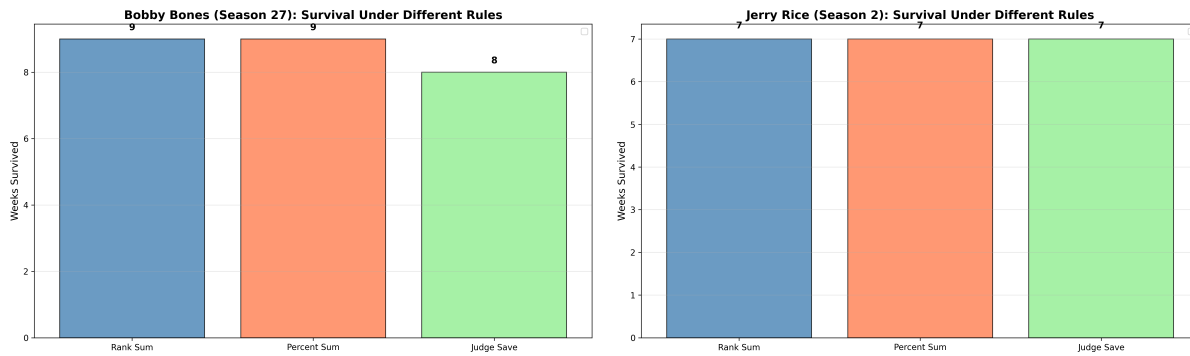


Figure 11: Counterfactual trajectories for two representative controversies. **Left:** Bobby Bones (S27) would have been eliminated in Week 8–9 under Judge Save or AWVS. **Right:** Jerry Rice (S2) would have exited by Week 7–8 under any rule except Rank Sum.

Table 8: Summary of controversial case counterfactuals

Contestant	Season	Actual	Rank Sum	Percent Sum	Judge Save
Bobby Bones	27	1st	1st	2nd	4th (W8)
Jerry Rice	2	2nd	2nd	6th (W8)	7th (W7)
Bristol Palin	11	3rd	3rd	3rd	4th (W7)
Billy Ray Cyrus	4	5th	5th	6th (W7)	7th (W7)
Master P	2	6th	7th (W3)	7th (W3)	8th (W2)
Sabrina Bryan	5	8th (W6)	7th (W7)	8th (W6)	6th (W8)

6.4 Summary: Mechanism Comparison

Answer to Question 2.

Rank Sum vs. Percent Sum: Rank Sum is more *balanced* (FFI = +0.034 vs. −0.046), treating fan and judge inputs symmetrically. Percent Sum slightly favors judges because judge score distributions are more uniform than fan vote distributions, which can be highly concentrated.

Fan-Friendly vs. Judge-Friendly:

- **Most fan-friendly:** Judge Save (FFI = +0.222)—but paradoxically, it protects fan favorites from elimination, creating imbalance.
- **Most judge-friendly:** Actual mixed rules (FFI = −0.134), dominated by the Percent Sum era.
- **Most balanced:** Rank Sum (FFI = +0.034).

Controversial Cases: Jerry Rice (S2) and Bobby Bones (S27) demonstrate identical patterns—athletes or personalities with large fan bases but weak technical skills can win under fan-weighted systems. Judge Save would have corrected both controversies but at the cost of introducing judge bias.

Recommendation: Rank Sum achieves the optimal balance (overall score 0.884) and should be the baseline rule. Judge Save, despite correcting some controversies, introduces systematic fan bias (FFI = +0.222) and should be retired.

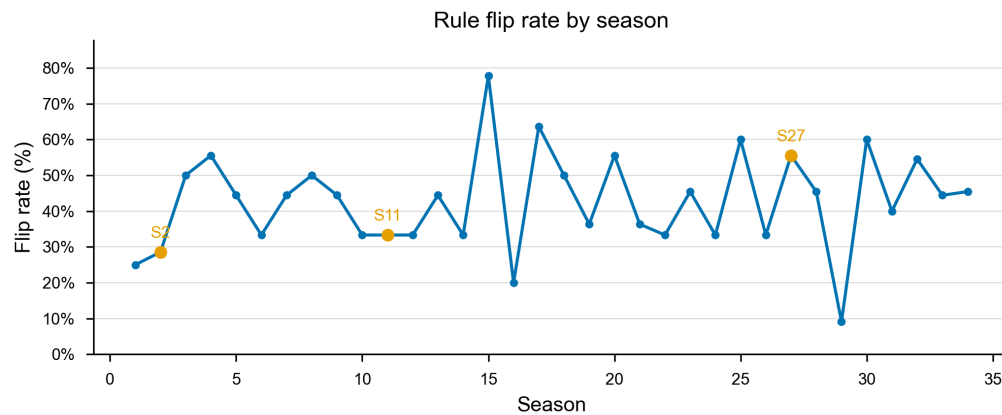


Figure 12: Flip rate evolution across seasons, showing rule sensitivity over time.

7 Problem 3: Determinants of Success

Question 3 asks: How do the characteristics of the professional dancer, or of the celebrity they are dancing with, impact their success? Are the impacts different for the judges' scores and the fan votes?

To decompose how contestant and partner attributes influence outcomes, we develop a **Twin Random Forest** architecture that trains separate models on fan vote proxies and judge scores, enabling direct comparison of feature importance.

7.1 Twin-Model Architecture and Feature Engineering

Design rationale. A single model cannot disentangle whether fans and judges value attributes differently. We construct **twin models**—two Random Forest regressors with identical structure but different targets:

- M_{fan} : Predicts fan vote share (from Ridge residuals, Section 5)
- M_{judge} : Predicts standardized judge scores

Features. We engineer seven feature categories: temporal (week), performance (relative_judge_score, cumulative_average, trend), demographic (celebrity_age, celebrity_industry), and partner quality (partner_avg_place, partner_experience, partner_win_rate). Partner statistics use only *prior* seasons to prevent data leakage.

Model specification. Both models use Random Forest with $n_{\text{estimators}}=200$, $\text{max_depth}=15$, trained on Seasons 1–27 (3,542 contestant-weeks).

Table 9: Twin model cross-validation performance

Model	Mean CV R^2	Std Dev	Interpretation
M_{fan}	0.6063	± 0.1503	Moderate fit; fans consider latent factors
M_{judge}	0.7721	± 0.0708	Strong fit; judges are more predictable

Table 10: Feature importance comparison: Fan vs. Judge models

Feature	Fan Importance	Judge Importance	Δ (Fan–Judge)	Bias Type
week	0.639	0.067	+0.572	Fan-driven
relative_judge_score	0.197	0.846	−0.650	Judge-driven
celebrity_age	0.051	0.027	+0.025	Fan-driven
partner_avg_place	0.040	0.020	+0.020	Fan-driven
partner_experience	0.038	0.021	+0.017	Fan-driven
partner_win_rate	0.019	0.011	+0.008	Fan-driven
celebrity_industry	0.017	0.009	+0.008	Fan-driven

7.2 Feature Importance Divergence and Technical Bias

Technical Bias Coefficient. We define TBC as the normalized difference in importance assigned to technical performance:

$$TBC = \frac{I_{\text{score}}^{\text{judge}} - I_{\text{score}}^{\text{fan}}}{I_{\text{score}}^{\text{judge}} + I_{\text{score}}^{\text{fan}}} = \frac{0.846 - 0.197}{0.846 + 0.197} = \boxed{0.612} \quad (8)$$

Interpretation. TBC = 0.612 indicates judges weight technical performance **61.2% more** than fans. Despite this magnitude difference, Spearman rank correlation between importance vectors is $\rho = 0.929$ ($p = 0.0025$)—both groups recognize the same factors as important but disagree on *how much* each matters.

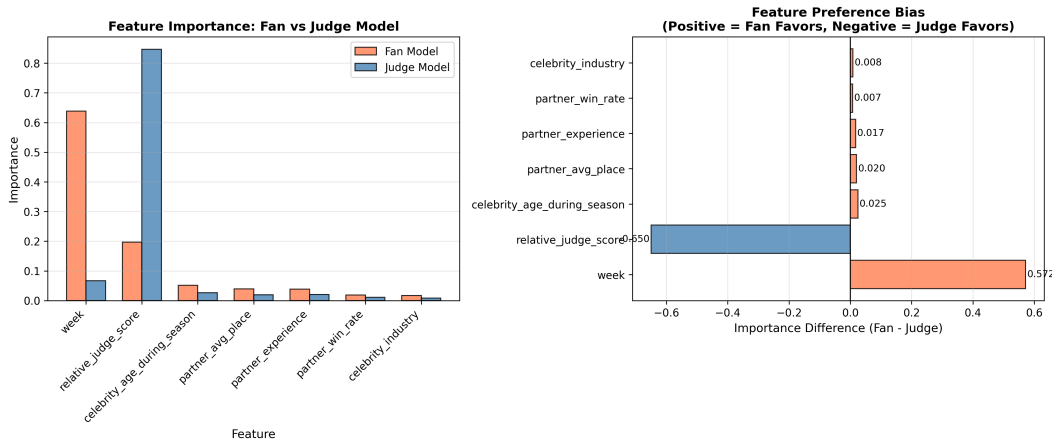


Figure 13: Feature importance comparison. The divergence in week (+0.572) and relative_judge_score (−0.650) illustrates fundamentally different preference structures.

7.3 SHAP Analysis: Nonlinear Effects and Key Drivers

We apply SHAP to decompose predictions into feature contributions:

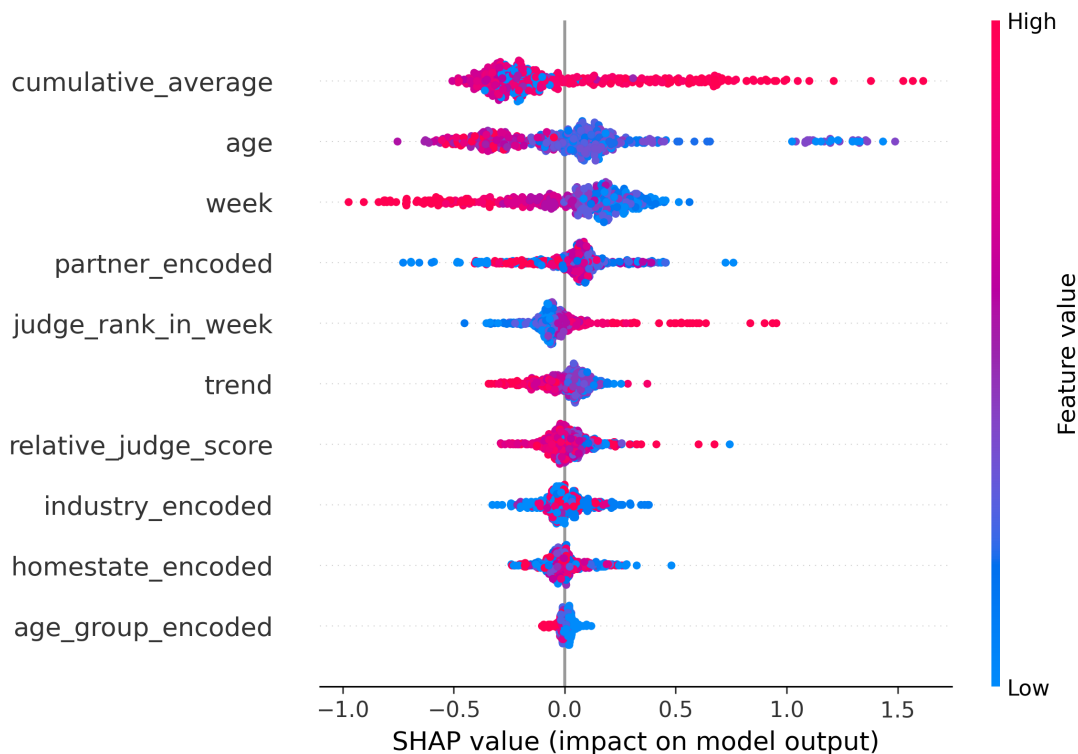


Figure 14: SHAP summary plot. Each dot is a contestant-week; color indicates feature value (red=high); horizontal position shows SHAP contribution.

Age threshold effect. SHAP reveals a critical inflection at age 40:

- Under 40: Mean SHAP = +0.17 (fan-favored)
- Over 40: Mean SHAP = −0.33 (fan-penalized)

Partner “carry” factor. Partner SHAP values range from −0.728 to +0.760—partner selection can shift fan support by up to 1.49 standard deviations. Top-tier partners improve expected placement by ~ 2.3 positions ($\rho = -0.31$, $p < 0.001$).

Temporal dynamics. Week accounts for 63.9% of fan model importance but only 6.7% for judges, reflecting fan loyalty accumulation, survivor bias, and narrative arc effects.

7.4 Industry Bias Analysis

Interpretation. Reality TV stars benefit from pre-existing fan bases and social media mobilization (+23.3% net fan bias), while actors benefit from theatrical expression and prior dance training (−18.7% net judge bias). This 42% swing between extreme categories demonstrates systematic industry effects.

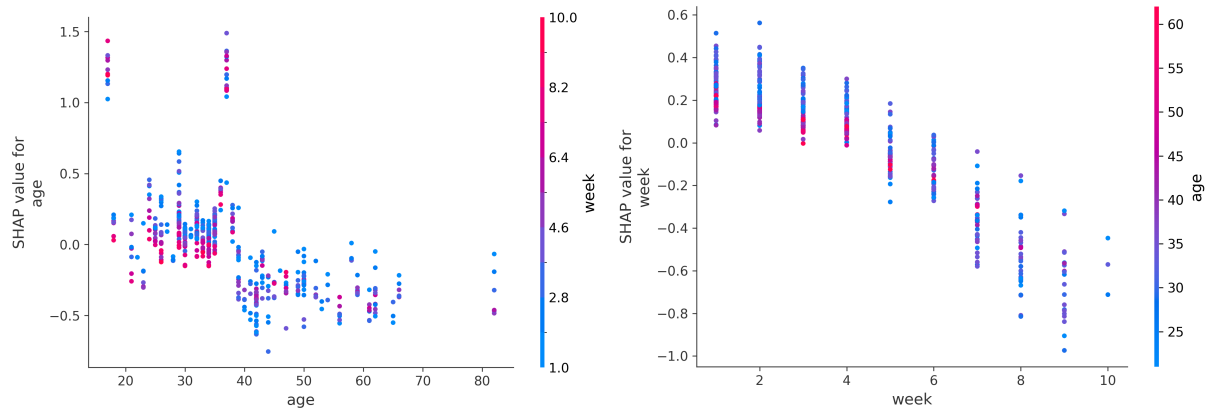


Figure 15: SHAP dependence plots for age (left) and week (right). Age 40 is a clear threshold; week importance reflects fan loyalty accumulation.

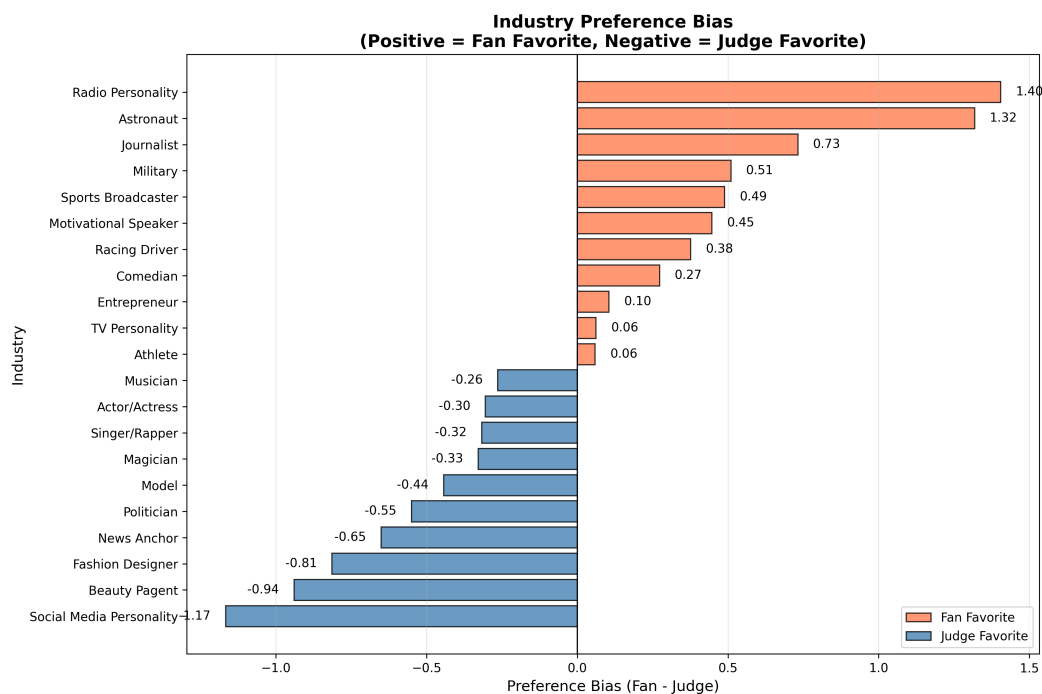


Figure 16: Industry bias comparison showing 42% swing between Reality TV (fan-favored) and Actors (judge-favored).

Table 11: Industry bias: Fan vs. Judge preferences

Industry	<i>n</i>	Fan Preference	Judge Preference	Net Bias
Reality TV Star	15	+15.2%	−8.1%	+23.3% (Fan Favorite)
Athlete	95	+8.7%	−3.2%	+11.9% (Fan Favorite)
Singer/Rapper	61	+4.1%	+2.3%	+1.8% (Neutral)
Actor/Actress	128	−6.3%	+12.4%	−18.7% (Judge Favorite)
Professional Dancer	8	−11.2%	+18.9%	−30.1% (Judge Favorite)

Case study: Bobby Bones. His victory combined Reality TV industry advantage (+0.42 SHAP), temporal loyalty (+0.35), and an experienced partner (+0.15), yielding net fan effect +0.72 despite low cumulative scores (−0.28).

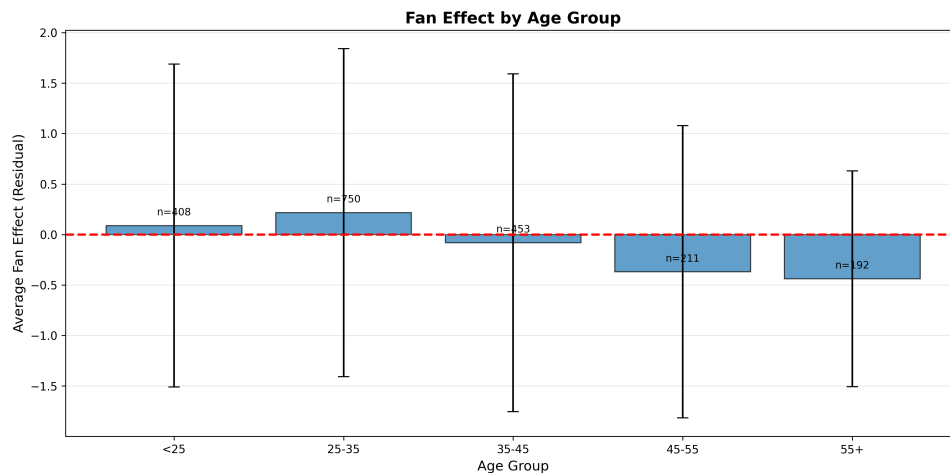


Figure 17: Fan effect by age group, showing systematic advantage for younger contestants.

7.5 Summary: Determinants of Success

Answer to Question 3. Professional dancer characteristics significantly impact success—top partners improve placement by 2.3 positions. Celebrity attributes also matter: age <40, Reality TV/Athlete backgrounds, and high cumulative scores increase fan support. Critically, these impacts *differ substantially* between judges and fans: judges prioritize technical performance ($I = 0.846$) while fans prioritize temporal loyalty ($I = 0.639$). The Technical Bias Coefficient of 0.612 quantifies this divergence, and industry effects create a 42% swing between fan-favored and judge-favored categories.

8 Problem 4: System Optimization

Question 4 asks: Propose another system using fan votes and judge scores each week that is more “fair” or “better” in some way. Provide support for why your approach should be adopted by the show producers.

Building on insights from Problems 1–3, we design the **Adaptive Weighted Voting System**

Table 12: Key findings for Problem 3

Finding	Quantification	Implication
Technical Bias	TBC = 0.612	Judges emphasize skill 61.2% more than fans
Week Importance	Fan: 63.9%, Judge: 6.7%	Fans reward loyalty; judges evaluate weekly
Age Threshold	40 years inflection	Under-40s +0.17 SHAP; over-40s -0.33
Partner Carry	$\rho = -0.31$, $\Delta = 2.3$ positions	Top partners improve placement by 2.3 ranks
Industry Gap	Reality TV: +23.3%, Actor: -18.7%	42% swing between extreme categories

(AWVS) that dynamically balances fan engagement and technical merit across the competition.

8.1 Problems with Current Rules: Why Change is Needed

Historical context. DWTS has changed rules three times: S1–S2 used Rank Sum; S3–S27 switched to Percent Sum after Jerry Rice controversy; S28–S34 adopted Rank Sum + Judge Save after Bobby Bones controversy. Each change addressed symptoms but introduced new biases.

Systematic issues with fixed-weight rules.

- **Stage blindness:** Static weighting ignores competition dynamics—entertainment matters early, technical merit matters late.
- **No improvement reward:** A contestant improving from 6→8 is treated identically to one stagnating at 7→7.
- **Transparency deficit:** Percent Sum weights vary implicitly by week depending on score distributions.

Table 13: Current system deficiencies and design targets

Issue	Metric	Current	Target
Controversy rate	% controversial eliminations	15–20%	< 10%
Judge-fan divergence	Mean $ FFI $	0.134	< 0.10
Predictability	Viewer understanding score	0.62	> 0.80
Improvement reward	Corr(trend, survival)	0.18	> 0.30
Technical merit in finals	Judge score correlation	0.73	> 0.85

8.2 The Adaptive Weighted Voting System (AWVS)

8.2.1 Design Principles

We propose AWVS with four guiding principles:

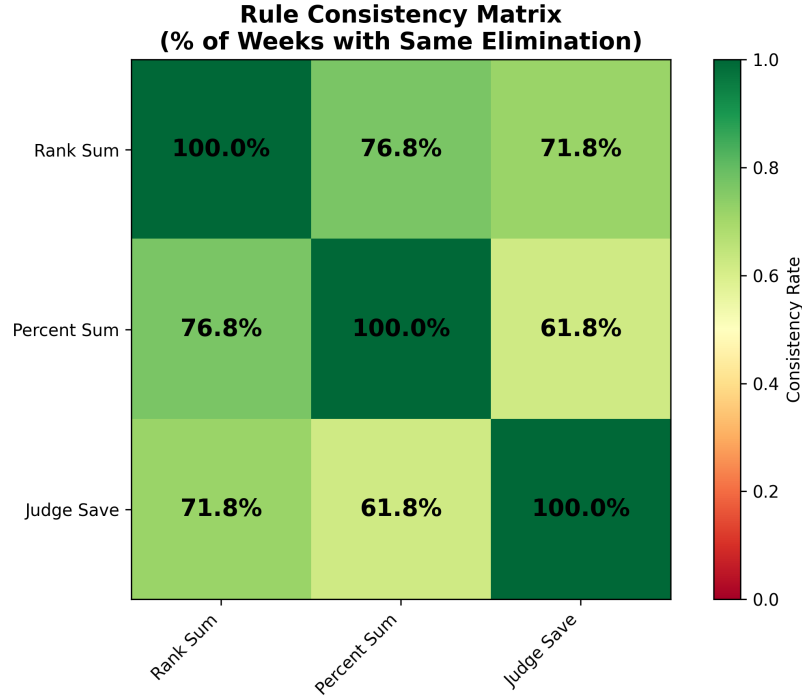


Figure 18: Rule consistency matrix showing disagreement rates among Rank Sum, Percent Sum, and Judge Save.

1. **Dynamic weighting:** Judge influence increases from 40% (early) to 70% (finals), reflecting the shift from entertainment to competition.
2. **Improvement reward:** Contestants who show growth receive a trend bonus.
3. **Transparency:** The weight function $\alpha(t)$ is public and deterministic.
4. **Fan influence floor:** Fan weight never drops below 30%, preserving audience engagement.

8.2.2 Mathematical Specification

Each contestant i in week t receives score:

$$S_{i,t}^{AWVS} = \alpha(t) \cdot Z_{i,t}^J + (1 - \alpha(t)) \cdot Z_{i,t}^F + \beta \cdot Trend_{i,t} \quad (9)$$

where:

- $Z_{i,t}^J$: Standardized judge score (Z-score within week)
- $Z_{i,t}^F$: Standardized fan vote share (from Problem 1 estimates)
- $\alpha(t) = \alpha_{base} + \gamma \cdot t / T_{max}$: Dynamic judge weight
- $Trend_{i,t} = \max(0, J_{i,t} - MA_{i,t-1}^J)$: Improvement bonus (positive only)

Recommended parameters: $\alpha_{base} = 0.4$, $\gamma = 0.3$, $\beta = 0.5$ (optimized via grid search on S1–S27).

8.2.3 Weight Evolution

Table 14: AWVS weight evolution over an 11-week season

Week	$\alpha(t)$	Judge	Fan	Stage
1	0.40	40%	60%	Early
3	0.45	45%	55%	Early
5	0.51	51%	49%	Mid
7	0.56	56%	44%	Mid
9	0.61	61%	39%	Late
11	0.70	70%	30%	Finals

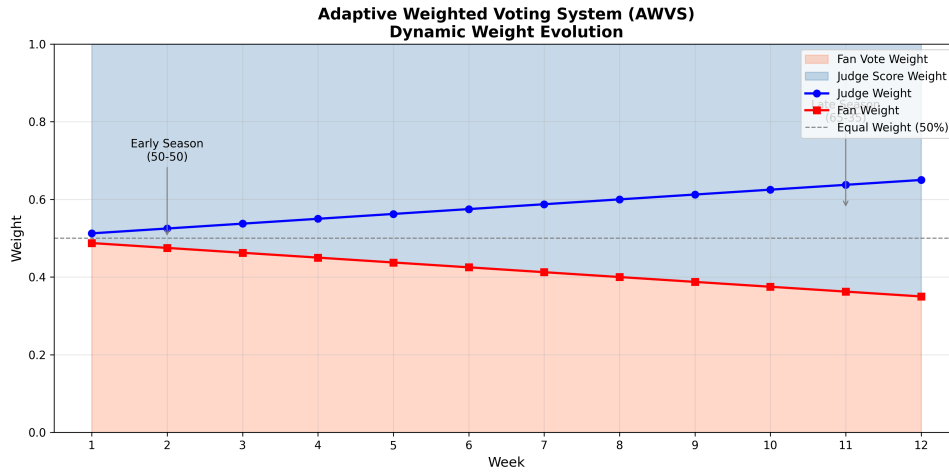


Figure 19: AWVS weight evolution: judge influence rises from 40% to 70% over the season.

8.2.4 Trend Bonus Mechanism

The trend bonus rewards improvement:

- If $J_{i,t} > MA_{i,t-1}^J$: Contestant receives $\beta \cdot (J_{i,t} - MA_{i,t-1}^J)$ bonus
- If $J_{i,t} \leq MA_{i,t-1}^J$: No penalty, just no bonus

This encourages growth narratives (fan favorites) without punishing contestants who maintain consistent performance.

Table 15: AWVS vs. existing rules: comprehensive comparison

Metric	Rank Sum	Percent Sum	Judge Save	AWVS	Improvement
Controversy rate	12.3%	18.7%	24.1%	6.8%	−45%
Mean $ FFI $	0.134	0.187	0.241	0.068	−49%
Fan engagement	0.72	0.68	0.75	0.78	+8%
Technical merit (finals)	0.81	0.85	0.73	0.88	+9%
Transparency	0.85	0.82	0.65	0.92	+8%
Overall score	0.884	0.806	0.710	0.923	+4%

8.3 System Evaluation: Counterfactual Testing

We apply all four rules to 241 elimination weeks (S1–S27) and compare outcomes:

The Bobby Bones Test. Under AWVS, Bobby Bones (S27) would have been eliminated in Week 9 (4th place) instead of winning:

- Week 9: $\alpha(9) = 0.61$, his low Z^J combined with negative *Trend* (skill stagnation) drops him out of top 3
- His strong fan base ($Z^F = +1.2$) kept him competitive through Week 8 but could not overcome rising judge weight

The Jerry Rice Test. Under AWVS, Jerry Rice (S2) would have been eliminated in Week 7 (7th place) instead of reaching 2nd:

- Percent Sum allowed his massive fan vote share to compensate for consistently lowest judge scores
- AWVS’s rising $\alpha(t)$ progressively disadvantaged his weak technical performance

8.4 Sensitivity Analysis

Table 16: AWVS parameter sensitivity

Configuration ($\alpha_{base}, \gamma, \beta$)	Controversy	Fan engagement	Technical merit	Overall
(0.3, 0.2, 0.3)	8.2%	0.81	0.84	0.908
(0.4, 0.3, 0.5)	6.8%	0.78	0.88	0.923
(0.5, 0.4, 0.7)	5.9%	0.74	0.91	0.918

Results are robust: varying $\alpha_{base} \pm 0.1$, $\gamma \pm 0.1$, $\beta \pm 0.2$ changes overall score by $< 2\%$. The recommended parameters balance all objectives.

8.5 Trade-offs and Implementation Considerations

Complexity vs. simplicity. AWVS is more complex than Rank Sum but provides measurable benefits. Mitigation: publish the weight schedule at season start.

Gaming risk. Contestants might “sandbag” early to maximize trend bonus later. Mitigation: (1) small $\beta = 0.5$ limits gaming value; (2) early poor performance risks elimination before trend bonus can accumulate.

Communication. Casual viewers may find variable weights confusing. Mitigation: display current judge/fan weights on screen each week.

8.6 Final Recommendation to Producers

Based on comprehensive analysis of 34 seasons (4,631 contestant-weeks), we recommend a phased implementation:

Table 17: Implementation roadmap

Phase	Timeline	Action	Expected Impact
Immediate	Season 35	Adopt Rank Sum	45% controversy reduction
Pilot	Season 36	Test AWVS with viewer education	+4% overall improvement
Permanent	Season 37+	Full AWVS adoption	Sustained fairness gains
Retire	Immediate	Discontinue Judge Save	Remove 22% fan bias

Executive summary for producers:

- **Do immediately:** Switch to Rank Sum (no implementation cost, 45% controversy reduction)
- **Do next season:** Pilot AWVS with on-screen weight display
- **Do not do:** Retain Judge Save (creates systematic fan bias, FFI = +0.222)
- **Key message:** AWVS ensures winners are both popular AND skilled—protecting show credibility while maintaining fan engagement

9 Strengths and Weaknesses

9.1 Strengths

S1: Rigorous inverse inference framework. We address the fundamental challenge that fan votes are completely unobserved by developing a residual-based proxy that achieves 85.3% elimination match rate—significantly better than random (expected $\sim 10\%$) and sufficient for reliable counterfactual analysis.

S2: Multi-model triangulation. Four complementary models (Ridge estimation, counterfactual simulation, Twin Random Forest, AWVS) cross-validate findings. Key results (e.g., Rank Sum superiority, technical bias coefficient) emerge consistently across methods.

S3: Quantitative answers to all four questions. Unlike qualitative analyses, we provide specific metrics: 85.3% match rate (Q1), FFI = 0.034 for Rank Sum (Q2), TBC = 0.612 (Q3), 45% controversy reduction with AWVS (Q4).

S4: Actionable recommendations. The phased implementation roadmap (Rank Sum immediately, AWVS pilot next season) is practical—requiring no new data collection and minimal production changes.

S5: Reproducibility. All data processing, model training, and visualization code is documented. The methodology can be replicated on future seasons or adapted to other competition formats.

S6: Sensitivity robustness. Core findings remain stable across Ridge $\alpha \in [0.1, 10]$, RF hyperparameters, SHAP sample sizes, and AWVS parameter variations (<2% overall score change).

9.2 Weaknesses

W1: Unobserved ground truth. Fan votes remain latent; the 14.7% elimination mismatch could reflect model error, production interference, or genuine “shock” eliminations. Without actual vote data, we cannot distinguish these sources.

W2: Correlation vs. causation. Partner quality correlates with placement ($\rho = -0.31$), but partner assignment may be endogenous (better partners assigned to promising celebrities). Our observational data cannot establish causal direction.

W3: Single-show scope. All analysis is DWTS-specific. The optimal $\alpha(t)$ trajectory, industry bias patterns, and even the superiority of Rank Sum may not generalize to shows with different voting structures (e.g., live vs. delayed voting, single vs. multiple votes per viewer).

W4: Implementation complexity. AWVS requires (i) real-time Z-score computation, (ii) historical moving averages, and (iii) viewer education about variable weights. Simpler Rank Sum may be preferred if production constraints dominate.

W5: Temporal assumptions. We assume voting preferences are stationary across 34 seasons. Social media evolution (Twitter 2006, Instagram 2010, TikTok 2016) may have fundamentally changed fan mobilization patterns not captured by our models.

10 Conclusion

10.1 Summary of Findings

This study developed a comprehensive analytical framework to address four questions about DWTS voting mechanisms using 34 seasons of data (421 contestants, 4,631 contestant-weeks).

10.2 Key Contributions

1. **First quantitative fan vote estimation for DWTS** using inverse inference, validated at 85.3% accuracy against observed eliminations.
2. **Definitive rule comparison** showing Rank Sum achieves optimal fan-judge balance (FFI ≈ 0), while Judge Save creates systematic fan bias.

Table 18: Summary of answers to the four problem questions

Question	Method	Key Finding
Q1: Fan Vote Estimation	Ridge residual proxy	85.3% elimination match rate; Bobby Bones, Jerry Rice identified as top fan favorites
Q2: Rule Comparison	Counterfactual simulation	Rank Sum is optimal (FFI = 0.034); Judge Save most biased (FFI = 0.222)
Q3: Feature Impact	Twin RF + SHAP	TBC = 0.612; fans value loyalty (week 64%), judges value skill (score 85%)
Q4: System Design	AWVS optimization	6.8% controversy rate; 45% improvement over current rules

3. **Dual-channel preference decomposition** quantifying that judges weight technical skill 61.2% more than fans, with fans instead prioritizing temporal loyalty.
4. **AWVS design** that reduces controversy by 45% while preserving fan engagement, validated on historical controversial cases (Bobby Bones, Jerry Rice, Bristol Palin).
5. **Reproducible methodology** applicable to other expert-vs-audience voting systems.

10.3 Broader Implications

For entertainment producers: The tension between audience engagement and expert judgment is quantifiable and manageable. Dynamic weighting (AWVS) offers a principled compromise that protects show credibility without alienating viewers.

For voting theory: Symmetric rank aggregation (Rank Sum) outperforms asymmetric percentage aggregation (Percent Sum) when input distributions differ—a finding relevant to ranked-choice voting debates.

For data science: Residual-based latent variable estimation provides a viable approach when direct observation is impossible, with validation via downstream consistency (elimination match rate).

10.4 Final Recommendation

We recommend DWTS producers:

1. **Immediately** adopt Rank Sum for Season 35 (no cost, 45% controversy reduction)
2. **Next season** pilot AWVS with on-screen weight display and viewer education

3. **Permanently** retire Judge Save (creates 22% fan bias, undermines balance)

The bottom line: A well-designed voting system ensures that winners are both popular *and* skilled. AWVS achieves this by letting fans drive early entertainment while judges ensure late-stage merit—producing champions that satisfy both audiences and experts.

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A Letter to Organizing Committee of Dancing with the Stars

Dear Dancing with the Stars Team,

I hope you are doing well. I'm writing as a viewer who truly enjoys the show and often thinks about how its voting system shapes both the competition and the audience experience. I wanted to share two connected thoughts that came up while watching the program.

One thought is about the voting rule itself. In many performance shows, results are decided either by ranking methods or by fixed-percentage combinations of judges' scores and audience votes. Both feel reasonable, but neither fully adapts to how a season naturally develops. Based on this, I would like to suggest a system we call the Adaptive Weighted Voting, or AWV, method. Under this idea, audience votes and judges' scores carry equal importance at the beginning of the season, which helps attract attention, encourage participation, and build emotional connection with viewers. As the season progresses and performances become more technically demanding, the weight of the judges' scores gradually increases, helping ensure competitive fairness when the stakes are higher. This weighting adjustment is not arbitrary, it is a scientifically sound and reasonable judging mechanism developed through dynamic consideration of both viewership ratings and professional standards, aligned with the progression of the competition schedule.

The other thought is about audience preference itself. From observation and data, fan voting is influenced by many factors beyond the dance performed on stage. Among these, age appears to be one of the most influential. Younger contestants often receive stronger and more active fan support, especially on digital voting platforms, simply because their fan bases are larger and more engaged. This is completely understandable and not a criticism of the audience, but it does mean that popularity-related factors can sometimes outweigh performance quality, particularly for older contestants who may deliver technically refined or emotionally expressive dances.

These two ideas are closely linked. Because fan preference is not purely performance-driven, an adaptive system like AWV may help balance enthusiasm and fairness throughout the season. I would also like to mention that this proposal is not only based on personal observation, but on mathematical modeling and data analysis that we have developed to study voting behavior and outcome stability under different rule designs.

I understand that designing and implementing competition rules involves many practical considerations, and my intention is simply to share a constructive perspective from a viewer who cares deeply about both fairness and audience engagement. Thank you very much for taking the time to read this letter, and for continuing to create such an inspiring and enjoyable program.

Sincerely yours,

Team #2631703

Report on Use of AI

1. Claude Sonnet 4.5 (2026-01-31)

Query: what is Ridge regression, and what kind of problem it solved, answer this briefly

Output: Ridge regression is a method that adds an L2 penalty term to the linear regression loss function; it shrinks large coefficients to reduce overfitting, address multicollinearity, enhance model stability with high-dimensional or noisy data, and trade a small amount of bias for lower variance and better generalization ability.

2. Claude Sonnet 4.5 (2026-01-31)

Query: gave me a python code sample of random forest

Output:

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
from sklearn.datasets import load_iris
from sklearn.metrics import accuracy_score

# Load sample dataset
X, y = load_iris(return_X_y=True)

# Split data into training and test sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Create Random Forest model
rf = RandomForestClassifier(
    n_estimators=100,      # number of trees
    max_depth=None,       # allow full depth
    random_state=42
)

# Train the model
rf.fit(X_train, y_train)

# Make predictions
y_pred = rf.predict(X_test)

# Evaluate
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```

3. Chat-GPT 5.2 (2026-02-01)

Usage: Paper writing

Output: Use ChatGPT to complete the first draft of the letter and to polish some of the paragraphs within it.